BLENDER ADD-ON · FREE AND OPEN SOURCE

Polyshell Builder

Turn a low-poly mesh into printable panels that glue together at exactly the right angles.



An 800 mm squirrel as 276 glue-up panels

What it does

Polyshell Builder turns a closed low-poly model into a kit of flat panels you can 3D-print and glue into a large, hollow sculpture. Every face becomes a thin shell with mitered edges, so neighbouring panels meet at exactly the angle of the original model.

Printed keys and cradles line up each joint, every panel carries its own ID and the IDs of its neighbours, and the whole kit comes out nested on print beds with a PDF guide for putting it together.

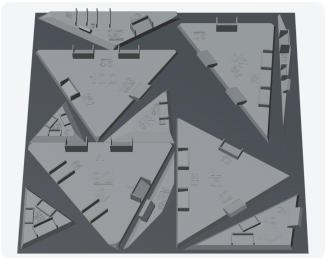
276	38	346	30 s
panels	print plates	keyed joints of 414	to build and export

The squirrel above: 800 mm tall, 6.4 mm skin, 256 × 256 mm print bed.

How it works

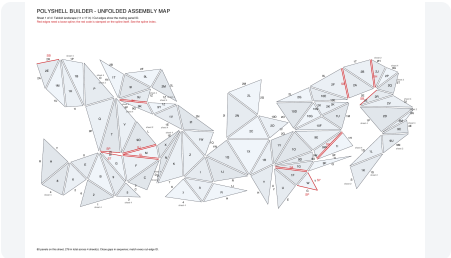
1. **Pick a mesh**
   Select any closed low-poly model in Blender.
2. **Set the size**
   Choose the finished size and a skin thickness.
3. **Describe your printer**
   Nozzle, layer height, fit and bed. Everything else is worked out for you.
4. **Build**
   One click makes the panels, keys, labels, print plates, STL files and PDF guide.
5. **Print and glue**
   Print outside-face down with no supports, then follow the map.

What you get



Print-ready STL plates

Panels nested on your bed size, outside face down, keys and labels up. One file per plate.



Assembly guide (PDF)

Labelled exploded views, an unfolded map of every seam in assembly order, and a spline index.

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Polyshell Builder 1.0.0  Complete:
276 panels, 46 splines, 38 plates

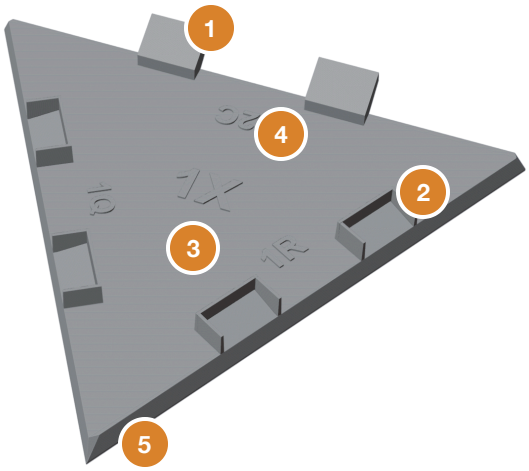
Thickness  6.40 mm (Standard)
Printer    0.40 mm nozzle, 0.20 mm layers
Key        17.6 x 9.0 x 4.0 mm
Print bed  256 x 256 mm

Edges 414: 346 keyed, 34 with splines,
34 without alignment
```

Build report

Every setting used and every edge or label that needs a second look, with the reason.

Inside every panel



1. **Printed key.** Reaches across the joint into the neighbour's cradle, locking position and angle.
2. **Cradle.** An open pocket that receives the neighbour's key with a fit you choose: snug, standard or loose.
3. **Panel ID.** Raised on the hidden inside face, so the outside stays clean.
4. **Neighbour IDs.** Every edge names the panel it joins; matching pairs glue together.
5. **Mitered edge.** The glue face lies exactly on the plane that halves the fold, so the angle is set by the part itself.

Built for accurate glue-ups

Checked in assembly
Every key, cradle, spline and label is tested against every other part in its final position before it is kept.

Hidden overlaps removed
Where many panels meet at a corner, their hidden inside corners are trimmed so they cannot collide.

Oversize faces split
Faces too big for your bed are divided automatically, with room left for their keys.

Splines for inward folds
Where no printed key can reach, a loose spline with a fixed bend sets the angle. Splines sharing a code are interchangeable.

Tuned from your printer
Clearances, key sizes, label sizes and reliefs are derived from nozzle, layer height and fit.

Labels you can read
Raised type sized to each panel, never mirrored; IDs such as 6 and 9 are underlined.



Splines print flat with no supports. Each is stamped with a short code, S1, S2 ... SA ..., that also appears beside the matching edge on both panels and in red on the unfolded map.

Choosing a thickness

PRESET	SKIN	BEST FOR
Economy	0.5% of size, 2–5 mm	Small pieces
Standard	0.8% of size, 3–8 mm	Most sculptures
Rigid	6.35 mm (¼ in)	Big pieces, stiffest joints

A thicker skin gives a wider glue face, which is what holds the angle. Panels are thinned automatically where sharp folds or thin parts of the model leave no room.

Printing and assembly

- Print each plate as exported: outside face on the bed, no supports.
- PLA and a cyanoacrylate (super) glue work well.
- Dry-fit a few neighbours first. Keys should slide in by hand; change the fit setting if not.
- Work outward from one panel using the map; red seams take the spline with the code shown.
- Check the report for edges without a key or spline, and glue those by eye.

What you need
- Blender 4.5 LTS or newer
- A closed (watertight) low-poly mesh, from a few dozen to a few hundred faces
- An FDM 3D printer; bed size is set in the add-on

Getting started
1. In Blender, open Edit › Preferences › Get Extensions and search for Polyshell Builder, or choose Install from Disk with the .zip.
2. Open the sidebar (N) and choose the Polyshell tab.
3. Select your model and click Build & Export Everything.

Free and open source under the GNU GPL v3 or later